Purdue University – UNP NS-12

Title:	Payload Interface Control Document		
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Release Table

Action	Team Member	Role	Date
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Change Log

Version	Date	Prepared By	Description
1.0	9/30/2025	Nick Babusis	Initial Release
2.0	9/16/2026	Alan Hsu	CDR Submission
2.1	10/22/2026	Alan Hsu	Minor Updates to reflect PIB Design Changes

Overview

The APEx payload subsystem consists of the systems required to operate and characterize the PMPD thruster.

Table 1: Payload Specifications

Bounding Volume	180mm (X) x 88mm (Y) x 120mm (Z)
Mass	1.4kg (TBR)
Typical Power Draw	15W (thrusting; TBR), 1W (standby; TBR)
Power interfaces	Raw Battery 12-16.8V 5V (TBD) 3.3V 12V (EGSE)
Electrical Connectors	1x 26-pin Harwin Gecko (Main Power/Data) 1x 4-pin Harwin M80 (EGSE)
Mounting Connections	Up to 20x screws on +X/-X panels
Fluid Connections	2x 10-32 Flat-bottom ports on propellant tank
Communication Protocols	SPI, I2C, GPIO Pulse Signal
Microcontroller	STM32G431 (TBR)
Sensor Channels	1x Flow meter (Frit) 1x Pressure/Temperature Sensor (Frit) 1x Pressure/Temperature Sensor (Tank) 2x Thermistor (Tank) 1x Thermistor (Discharge Power Supply) 1x Thermistor (Igniter Power Supply) 1x Thermistors (Payload Interface Board) 1x MCU Temperature 1x Igniter Peak Voltage 1x Igniter Peak Current (Hall effect) 1x Igniter Supply Voltage 1x Discharge Voltage 1x Discharge Current (Rogowski)
Actuators	2x Polyimide Heater (Tank) 3x Solenoid (Tank to Frit) 1x Igniter Switch

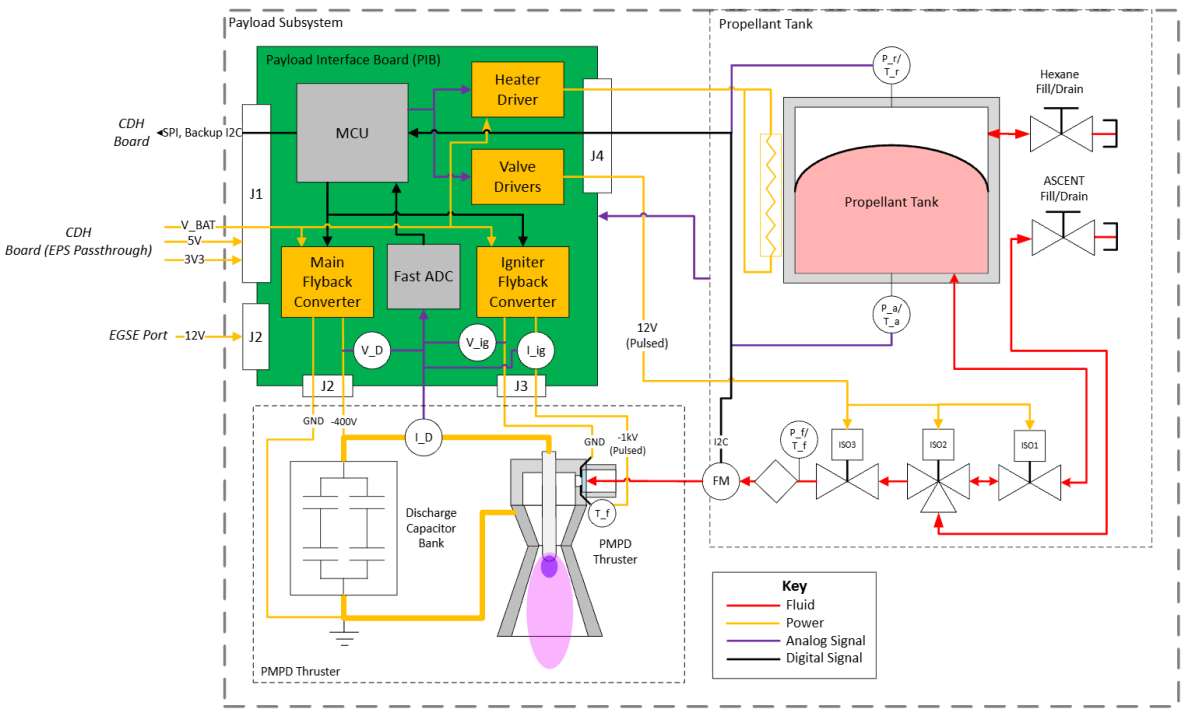


Figure 1: Full System Connection Diagram

Mechanical Interfaces

In terms of structural security, the payload subsystem is largely secured to the satellite via frame mounting, having multiple components within the subsystem mounted directly to said frame.

More specifically, the tank is mounted to an upper partition on the frame, which separates the payload from other subsystems. It is to be mounted via 4 screws on the top face of the panel extending into the inner walls of the tank, screwed in the +Z direction, as well as 2 screws on either side of the tank, screwed in the +Y/-Y direction. Said mounting methods are pictured below for reference.

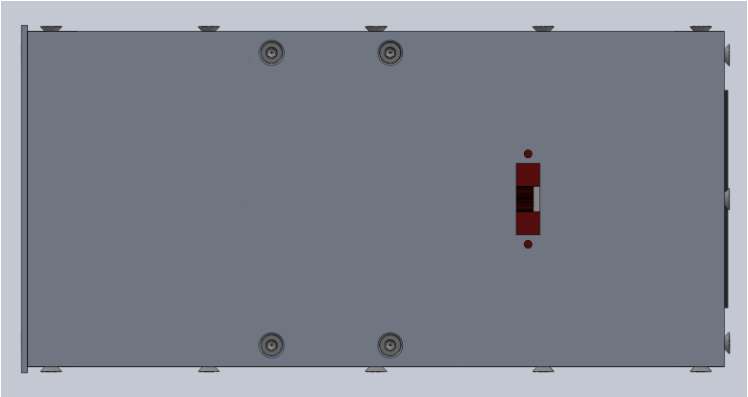


Figure 2: 4 Top Panel Tank Mounting Screws

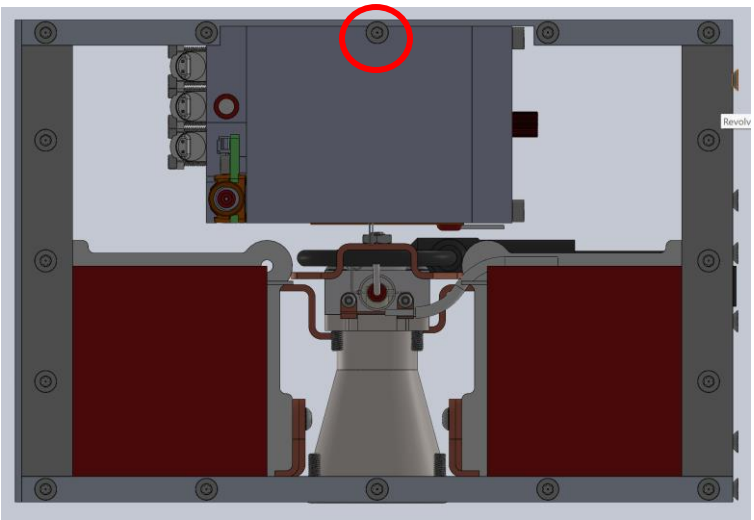


Figure 3: 2 Side Frame Tank Mounting Screws (Circled in Red & Mirrored on Other Side)

The capacitors are to be mounted to the frame as well, with the 2 of them being secured to the side panels of the frame, separating the inside of the satellite from the outside space. To be specific, each capacitor is secured via 4 screws on the nearest side panel, screwed in the +X/-X direction, Additionally, each capacitor has a bracket encasing it, which wrap around the capacitor and is secured to the side and bottom panels, via screwed in the +X/-X and –Z direction. Again, this system is pictured below for reference.

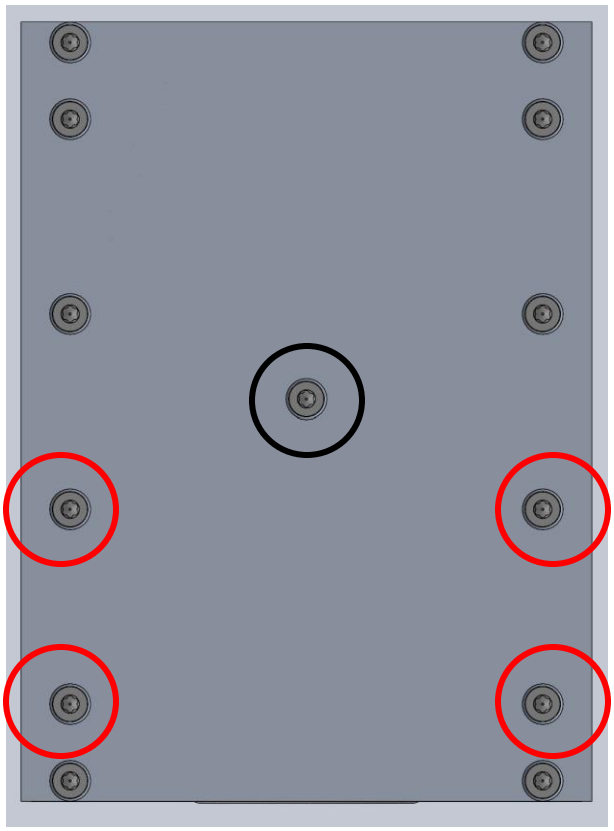


Figure 4: 2 Side Panel Capacitor Mounting and Bracket Screws (Capacitor Circled in Red, Bracket in Black & Mirrored on Other Side)

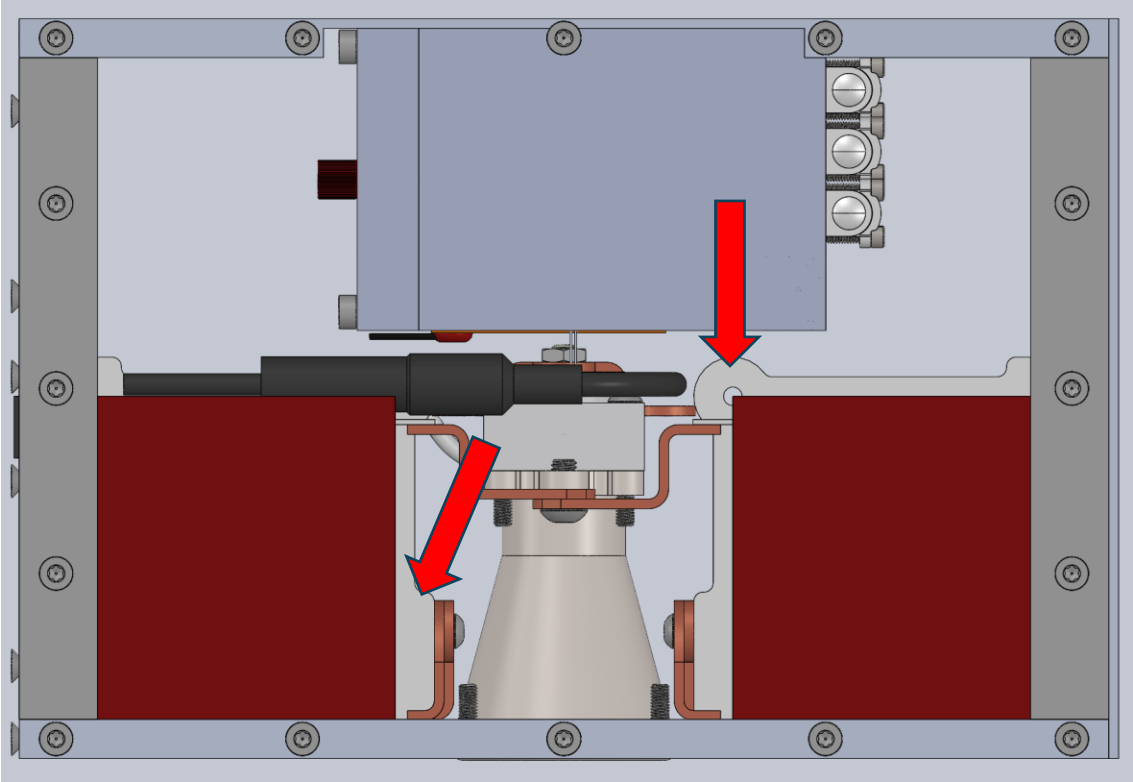


Figure 5: 2 Side View Capacitor Mounting and Bracket (Pointed to Via Red Arrow)

Additionally, the nozzle and accompanying frit are mounted to the capacitor securing brackets via screws in the +X/-X and +Z/-Z directions. That said, the nozzle specifically, is mounted to the bottom panel of the frame via 4 screws in the –Z direction, for extra security. Once more, images are provided for reference below.

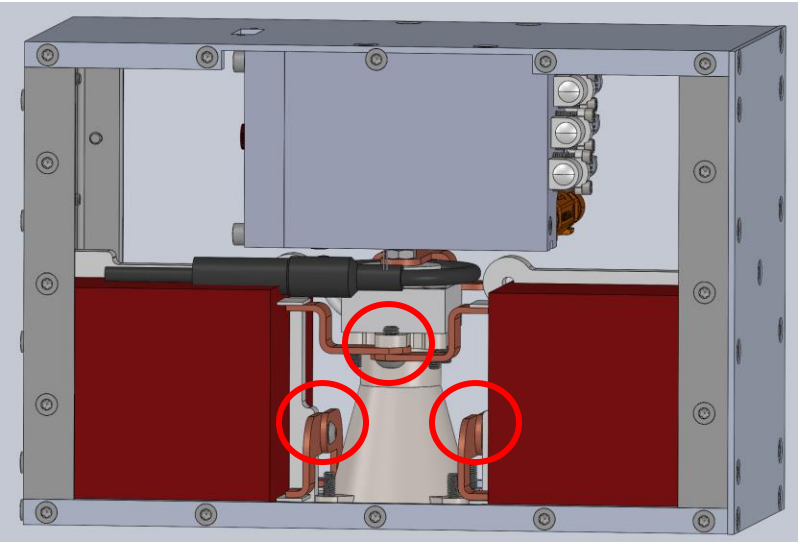


Figure 6: 2 Frit and Nozzle Bracket Securing (Circled in Red)

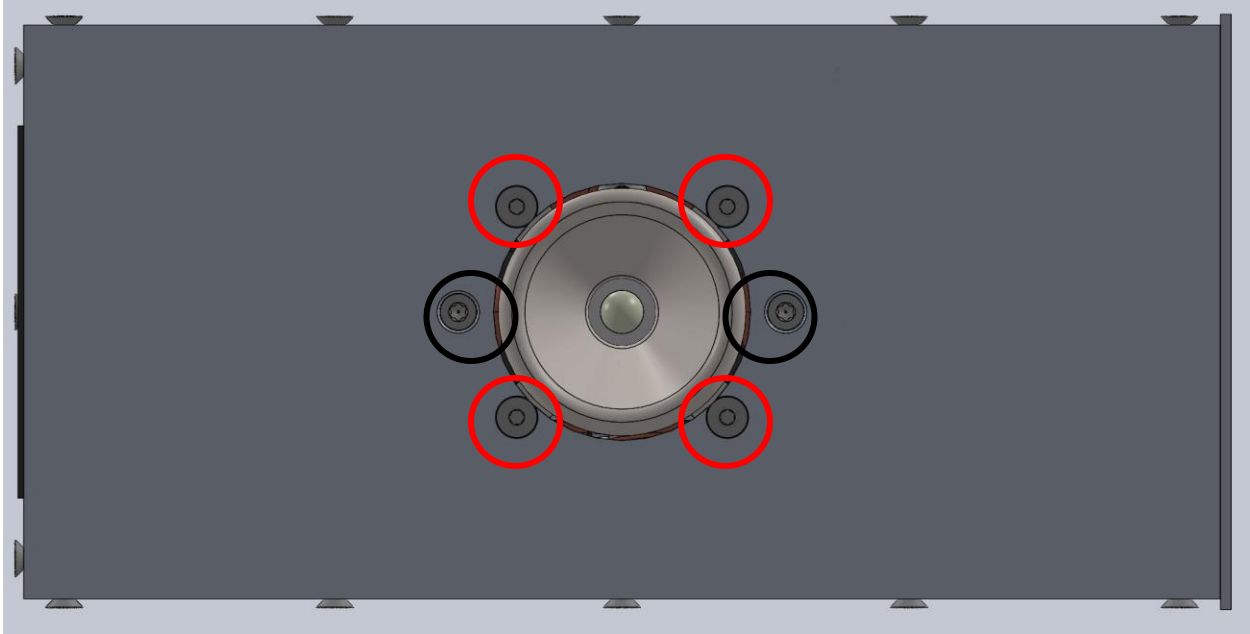


Figure 7: 2 Nozzle and Bracket Panel Securing Screws (Nozzle Circled in Red & Bracket in Black)

Electrical Interfaces

Pin	Func.	Pin	Func.
1	BAT	14	GND
2	BAT	15	GND
3	BAT	16	GND
4	3V3	17	GND
5	3V3	18	GND
6	SPI_MISO	19	GND
7	SPI_MOSI	20	I2C_SCL
8	SPI_SCK	21	I2C_SDA
9	SPI_CS	22	IGNITER
10	NC	23	DRDY
11	NC	24	ARM
12	5V	25	GND
13	5V	26	GND

Table 2: G125-MS12605M2P connector pinout

The Payload Interface Board (PIB) will expose a single G125 26 pin connector intended to interface with CDH. The G125-MS12605M2P connector has a current rating of 2.8A for an individual contact, and 2A per contact when multiple contacts are conducting. At 15W average and 12V (worst case) battery voltage three pins allocated to battery voltage means 0.42A per contact. This is ideal because the crimp of G125-MS12605M2P has a maximum wire gauge of 26 AWG, which is not well suited for high current. Redundancy is also achieved by using a minimum of two pins per power connection.

3.3V is used for powering the microcontroller, digital logic, and analog reference rails. 5V is used for gate drive voltage on power supply circuitry and is used in conjunction with an internally regulated –5V rail to power op amps for analog signal processing.

BAT is used to power the discharge and igniter supplies, the valves through an onboard 12V internal regulator, and the tank heaters.

Three latching current monitors with auto retry monitor the current drawn off of the BAT pins and trip at 3.6A for the entire payload, 3.2A for the discharge supply, and 0.5A for the igniter. Undervoltage lockout features disable the discharge and igniter supplies at 12.75V falling (typ) and disable the payload's battery rail as a whole at 12V falling (typ). Hysteresis

is described in Table 3. No latching current limiter or undervoltage lockout feature exists on the 3.3V or 5V rails, and it is assumed the EPS delivers the correct voltage and limits its output current accordingly for these rails. The majority of the electrical power while thrusting will be delivered through the BAT bus, with only a few watts delivered for logic on the 3.3V and 5V buses.

	min	typ	max
VBAT Main UVLO falling threshold (off)	11.82V	12.00V	12.18V
VBAT Main UVLO rising threshold (re-enable)	12.48V	12.72V	12.96V
VBAT Main OVLO rising threshold (off)	17.73V	18.00V	18.27V
VBAT Main OVLO falling threshold (re-enable)	17.10V	17.46V	17.82V
VBAT Discharge/Igniter UVLO falling threshold (off)	12.56V	12.75V	12.94V
VBAT Discharge/Igniter UVLO rising threshold (re-enable)	13.26V	13.51V	13.77V
VBAT Discharge/Igniter OVLO rising threshold (off)	17.72V	17.99V	18.26V
VBAT Discharge/Igniter OVLO falling threshold (re-enable)	17.09V	17.45V	17.81V
Main OCP trip threshold	3.24A	3.60A	3.95A
Discharge Supply trip threshold	2.86A	3.20A	3.53A
Igniter Supply trip threshold	0.47A	0.50A	0.54A
OCP Retry Delay		200ms	

Table 3: Battery Bus Internal Protection Setpoints

Additionally, onboard power circuitry will only pass BAT to the converters, valves, and heaters, if 3.3V is also powered and the microcontroller is healthy. Since both HV buses are independently controlled by their own load switch, each heater is independently controlled by its own MOSFET, and there is an additional load switch controlling power to the internal BAT bus, all hazardous conditions on these devices are protected against with tolerance to a FET failing short circuit.

Because system requirements call for launch inhibits to be placed on the EPS, inhibiting power to the rest of the system, it is assumed that these are in place and no electrical inhibits are provided on the Payload system. ARM/SAFE (described later) is provided as a

safety measure during ground testing and as prevention against single event effects causing spurious turn-on, but it is not designed to meet the two-fault tolerant requirement of UNP inhibit design.

Data Interfaces

PIB exposes a primary SPI connection to the MCU, which acts as an SPI slave. A backup I2C interface is provided should SPI fail but acts at a lower bitrate. Three digital outputs indicate when ignitor pulses happen at the hardware (IGNITER), when ADC data is collected and ready for retrieval by CDH (DRDY), and whether the payload is ARMED or SAFE (ARM). IGNITER is driven by a buffer from the igniter pulse commanded by the MCU on the PIB, so its length is short, in the single digit microseconds range.

PIB firmware stores only an ARM/SAFE state, which determines whether high voltage supplies are active and whether propellant solenoid valves are open. ARM serves to ensure safety on potential ground testing, to ensure spurious commands do not cause thrust, and to minimize the voltage stress applied to high voltage converters on orbit by keeping them cold outside of thrusting periods. PIB defaults to SAFE and ARMs only upon the correct SPI/I2C command from CDH. Lack of communication from CDH for a specified watchdog interval will result in PIB defaulting back to SAFE state automatically, in case of comms failure. PIB polls all sensors except waveform sensors for telemetry data at 10 Hz (TBR) frequency, and data is stored in RAM and overwritten every polling cycle. Waveform data is recorded on a per pulse basis and is overwritten each pulse with the assumption that CDH will retrieve necessary data between pulses. The intent is for CDH to handle all complex states and reliable data storage, and for PIB to only actuate the thruster and immediately return the data to CDH.

Intended CONOPS is as follows:

1. Before experiment start, CDH issues command over SPI/I2C to ARM the payload. PIB responds by opening the three solenoid valves and enabling the high voltage discharge and ignitor supplies. After basic health checks come back OK, PIB asserts the ARM flag.
2. During the experiment, CDH will issue a command to pulse the thruster over SPI/I2C approximately once every second with contents including ignitor pulse width as an argument, and payload will respond by activating the ignitor upon receipt of such command, and pulsing the IGNITOR signal with positive edge aligned with ignition.

3. Payload collects voltage/current waveform data from the pulse for 20us and asserts the DRDY flag to indicate to CDH that data is ready for retrieval.
4. CDH collects the data by SPI/I2C read command, requiring about 330 bytes per pulse. PIB de-asserts the DRDY flag upon data retrieval completion
    a. Data retrieval should take less than 1ms including overhead, assuming an SPI CLK frequency of 1MHz. This allows multiple attempts at data retrieval in case of failed transactions and ample time for non pulse related telemetry/command between pulses.
5. Repeat (2-4) until burn is complete, then CDH commands PIB to SAFE the payload, to which payload responds by switching off the high voltage rails and closing all valves, then de-asserting ARM
6. If SPI/I2C comms is lost mid-burn, software watchdog on PIB SAFEs the payload automatically after a set (on the order of 10s; TBR) timeout period without command.
7. Before, during, and after burn, CDH may issue commands for changing tank temperature setpoint, and PIB will regulate tank thermistor temperatures to these setpoints.
8. CDH may also telemeter flow rates, pressure/temperatures, and other housekeeping data not related to pulse waveforms at any point, regardless of DRDY flag. Retrieved data will be from the last time PIB polled sensors, which occurs at a TBR frequency loop.

See the Command & Telemetry List document for more details including a register map for commands accepted by PIB.

GSE Interfaces

Fill and drain of propellant and pressurant is done through 10-32 fittings on the propellant and pressurant sides of the tank. The +Y or –Y panel must be removed to access the tank within. 10-32 Flat-bottom ports on each side are nominally fitted with plugs, which can be removed and replaced with fittings to perform fill and drain.

Since fill/drain of propellant requires two of three solenoid valves to be open, separate circuitry on PIB opens these two specific valves upon command from the EGSE over a digital 3.3V logic level pin. The circuitry driving the valves from the EGSE port's power is separate from the spacecraft power and diode ORs into valve drivers shared with flight circuitry. Powering EGSE does not power any flight circuitry apart from the valve drivers, and EGSE circuitry remains unpowered in flight.

EGSE 12V MUST be supplied with at least 12V, and at most 13V. 12.5V is recommended to account for diode voltage drop from the diode OR for voltage driving. A UVLO feature on the EGSE circuitry does not allow driving of the valves if supply voltage is less than 9.6V and is guaranteed to allow operation above 11.8V. Driving the OPEN/CLOSE pin high opens the NC valve by applying 12V across it and opens the tank latching valve by pulsing it in one direction for 63 to 205ms. A falling edge on the OPEN/CLOSE pin closes the tank latching valve by pulsing it in the opposite direction for 63 to 205ms. Due to the length of these pulses, rising and falling transitions on the OPEN/CLOSE line must be longer than 205ms to ensure no overlap of commands, which would shorten pulses. Due to the startup transient behavior, a pulse is automatically transmitted to the latching valve corresponding to the state of the OPEN/CLOSE pin whenever power is first applied to the EGSE circuitry. Therefore it is required that the OPEN/CLOSE pin not be driven high before power is applied to the 12V pin to avoid an accidental OPEN command. Additionally, no pulse is automatically transmitted when 12V is removed from the EGSE connector, so the OPEN/CLOSE line MUST be brought low for at least 205ms before removing 12V power to ensure the latching valve closes.

The connector pinout is listed below in detail and is passed through the satellite chassis EGSE connector.

Pin	Func.	Pin	Func.
1	12V	3	GND
2	OPEN/CLOSE	4	GND

Table 4: M80-5100442 connector pinout

Fill drain is designed to occur while the satellite is powered off, with EPS inhibited from powering the satellite.

Thermal Interfaces

The payload primarily dissipates 5W (TBR) of heat in the cathode of the thruster, and this is transferred out through a Boron Nitride substrate to the titanium anode (nozzle) of the thruster and the copper conductors to the main discharge capacitors, both of which dissipate into the +Z face of the frame.

The payload interface board's high voltage converters are expected to dissipate (TBR) watts of switching inefficiency, which dissipate heat to an Aluminum contoured heatsink plate on the -X face of the tank assembly, dissipating heat both into the tank and into the –Z

panel of the PIB, which interfaces with the frame. This helps with maintaining the temperature of the propellant and pressurant in the tank.

The tank is maintained around 37˚C using polyamide heaters. The heaters are mounted on the +Z face of the tank assembly, and waste heat from the heaters radiates towards the +Z face and Boron Nitride substrate. The tank itself, made of Delrin plastic, is mounted to and dissipates heat to the –Z face of the frame.

TBD whether copper thermal straps are added to improve conduction to the frame. One risk item is the titanium selected for the anode, which has poor thermal conductivity compared to copper.

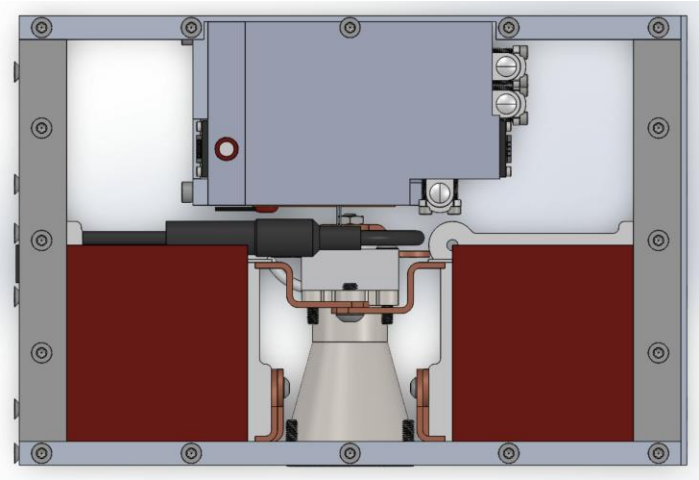


Figure 8: Payload Side View (TBR), note copper conductors connecting discharge capacitors and nozzle

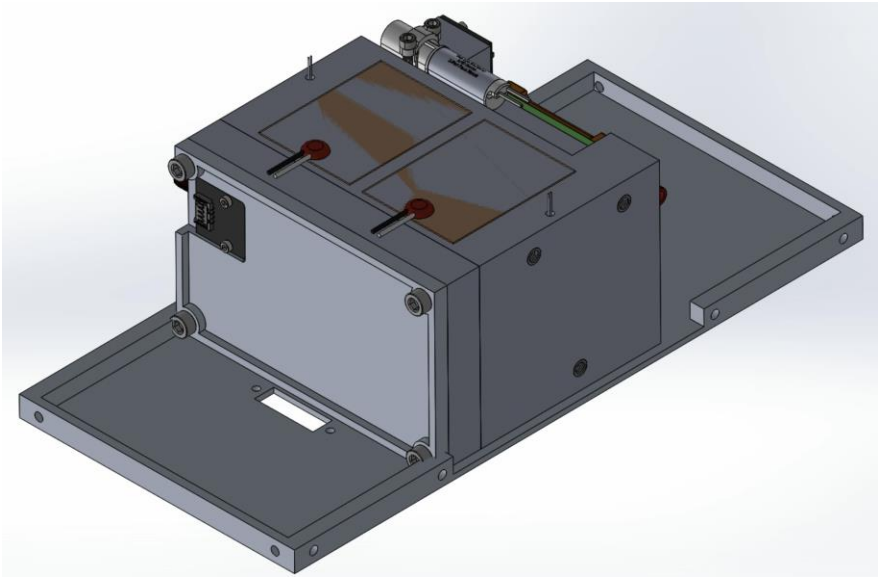


Figure 9: Tank assembly with –Z Panel